HANDWRITTEN STUDY NOTES

GEOSYNTHETICS AND REINFORCED EARTH STRUCTURES

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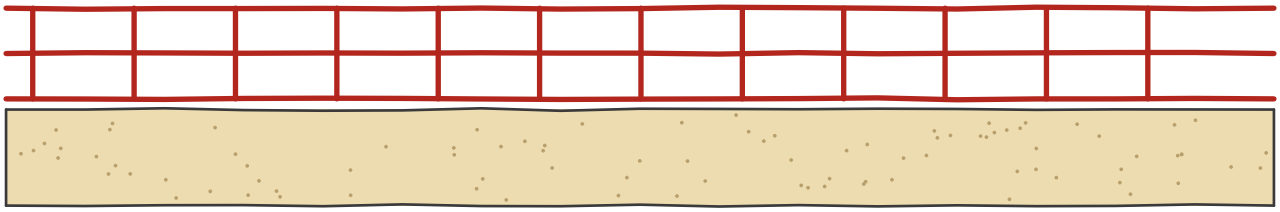
UNIT – I

Reinforced Earth

Syllabus topics covered in these notes:

- Reinforced Earth: concept, effects of reinforcement on soils
- Equal (enhanced) confining pressure and pseudo-cohesion concepts
- Materials; friction coefficient: definition, laboratory determination, factors affecting it
- Applications of reinforced earth

soil + reinforcement = strong composite



1. Reinforced Earth – Concept

Reinforced earth (Terre Armée) is a **composite material** made of compacted frictional soil + tension-resisting reinforcement (steel strips, geogrids, geotextiles) placed in layers. It was invented by the French architect-engineer **Henri Vidal** in the 1960s; the first structure (a wall) was built at Pragnères, France, in 1965. Today it is also called **MSE** (Mechanically Stabilised Earth).

- **1963–65:** Vidal patented the system; first wall at Pragnères (Pyrenees).
- **1972:** first wall in the USA (California, SR-39).
- **1970s–80s:** geotextile and then geogrid reinforcement introduced; now widely used worldwide, including India (flyover approaches, bridge abutments, hill roads).

Soil is strong in compression & shear but very weak in tension. Reinforcement is strong in tension. When combined, the soil mass behaves like a block of RCC: soil takes compression, reinforcement takes tension.

Remember

Analogy: RCC = concrete + steel bars

Reinforced earth = soil + strips / grids / sheets

Basic mechanism

- Soil mass tries to deform laterally when loaded (vertical load, self-weight, surcharge).
- Reinforcement is placed horizontally, in the direction of tensile strain, so soil tries to move relative to the strip.
- Friction (and passive bearing for grids) between soil and reinforcement develops **tension T** in the reinforcement.
- Lateral movement is restrained, so soil acts as if it were under higher confinement; the mass becomes **coherent**, stiff and strong.

Components of a reinforced earth structure

1. **Backfill** – compacted granular, free-draining frictional soil.

2. **Reinforcement** – metallic or polymeric tensile elements in horizontal layers.
3. **Facing** – precast concrete panels, modular blocks, wrap-around geosynthetic, gabion, welded wire. It only prevents local sloughing; it carries little load.
4. **Foundation / leveling pad** and **drainage** system.

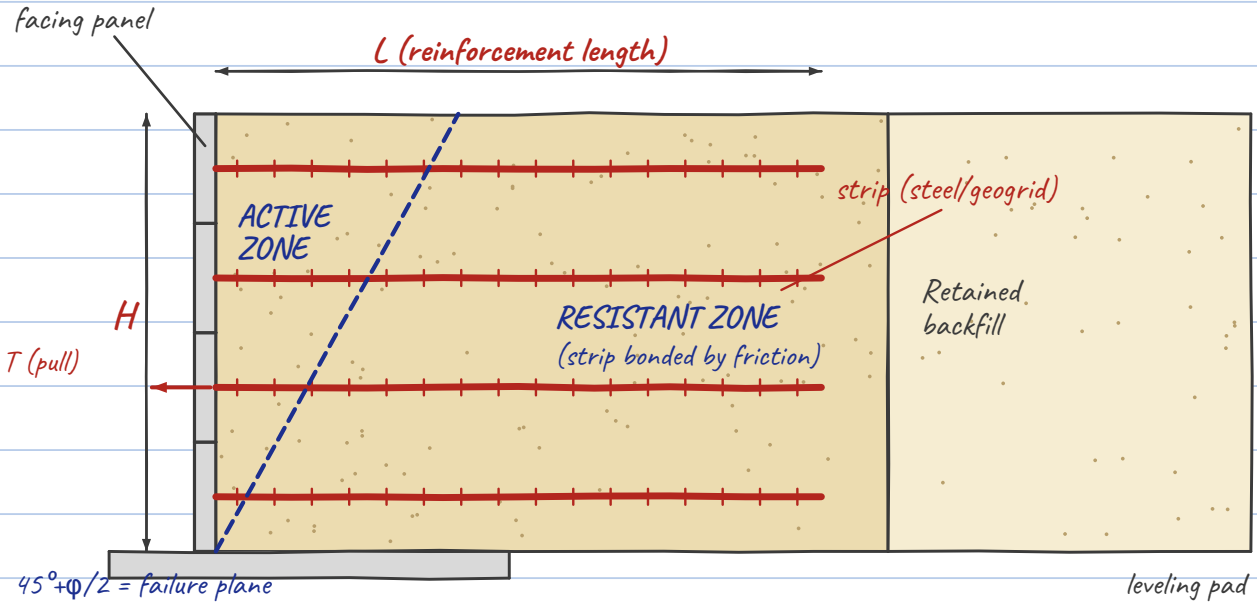


Fig 1.1 Reinforced earth wall – strips, facing and failure plane

Reinforced earth vs RCC

Property	RCC	Reinforced earth
Matrix	Concrete (cement bound)	Soil (friction bound)
Reinforcement	Steel bars, bond by adhesion	Strips / grids, bond by friction
Behaviour	Rigid, brittle	Flexible, ductile
Settlement	Cracks easily	Tolerates differential settlement

2. Effects of Reinforcement on Soils

- Increases **shear strength** and **bearing capacity** of the soil mass.
- Increases **stiffness** (modulus) and reduces **settlement**.
- Restrains **lateral strain** (lateral spreading) and horizontal deformations.
- Improves **ductility**: failure is gradual, not sudden; post-peak strength loss is reduced.

- Results in **anisotropic** strength: strong along reinforcement direction, weaker across it.
- Allows **steeper slopes** and vertical walls with granular soil.

Two classical concepts explain the strength gain (Yang 1972; Schlosser and Long 1972; Hausmann 1976):

(a) Equal (enhanced) confining pressure concept

Reinforcement restrains the lateral expansion of soil. This restraint is equivalent to an **additional confining pressure** $\Delta\sigma_3$ acting on the soil, without any change in its friction angle φ.

$$\Delta\sigma_3 = T / (S_v \cdot S_h) \text{ (strips of spacing } S_v \times S_h)$$
$$\Delta\sigma_3 = T / S_v \text{ (continuous sheets, } T \text{ per metre width)}$$

where T = tension mobilised in the reinforcement at failure (per strip / per metre).

At failure of the soil (Rankine/Mohr–Coulomb):

$$\sigma_1 = (\sigma_3 + \Delta\sigma_3) K_p, \quad K_p = \tan^2(45° + \varphi/2)$$
$$\Delta\sigma_1 = \Delta\sigma_3 \cdot K_p$$

So the Mohr circle becomes **bigger** for the same σ_3, but the failure envelope of the soil (same φ) is unchanged. Triaxial tests (Yang 1972) show: strength increases with number of layers, closer spacing and higher modulus of reinforcement.

(b) Pseudo-cohesion (apparent cohesion) concept

Here the reinforced soil is treated as a **new anisotropic material** that has the same φ but an **apparent cohesion** c_R – even for a cohesionless sand. The failure envelope shifts upward by c_R.

Derivation: for the reinforced soil $\sigma_1 = \sigma_3 K_p + 2c_R\sqrt{K_p}$. Equating with $\sigma_1 = \sigma_3 K_p + \Delta\sigma_3 K_p$ gives

$$c_R = (\Delta\sigma_3 / 2) \cdot \sqrt{K_p} = [T / (2 S_v S_h)] \cdot \tan(45° + \varphi/2)$$

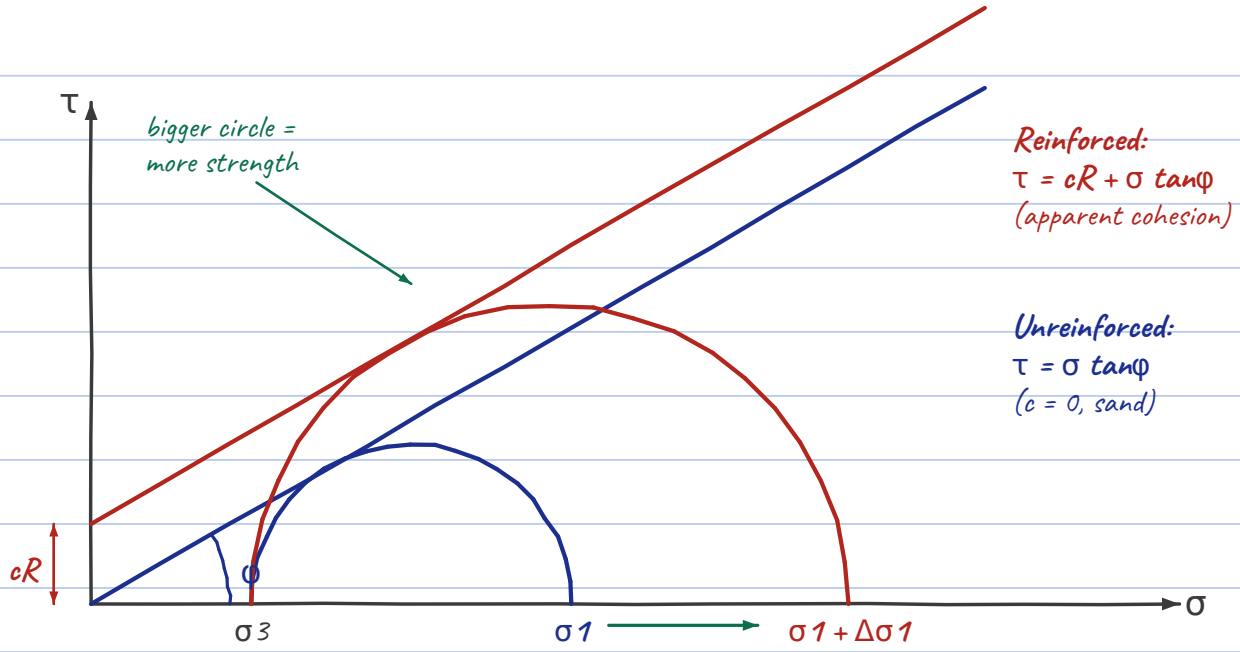


Fig 1.2 Mohr circles – effect of reinforcement

Example 1

Worked example: sand $\varphi = 30^\circ$, reinforcement tension $T = 15$ kN per strip, $S_v = 0.5$ m, $S_h = 0.75$ m.

$\Delta\sigma_3 = 15 / (0.5 \times 0.75) = 40$ kPa; $K_p = 3$; $\sqrt{K_p} = 1.732$

$c_R = (40 / 2) \times 1.732 = 34.6$ kPa; $\Delta\sigma_1 = 40 \times 3 = 120$ kPa

Comparison of the two concepts

Point	Enhanced confining pressure	Pseudo-cohesion
Idea	Extra σ3 on same soil	New material with cohesion c_R
φ of soil	Unchanged	Unchanged
Envelope	Same line; bigger circle	Line shifted up by c_R
Best for	Closely spaced sheets, triaxial	Strips, design of reinforced beds

3. Materials for Reinforced Earth

3.1 Backfill (reinforced fill)

- Must be **granular, free draining**, with good friction angle $\varphi \geq 30^\circ$–34°.
- Fines passing 75 μm sieve ≤ **15 %**; plasticity index PI ≤ **6**; maximum particle size ≤ 75 mm (to avoid damage to reinforcement).

- Compaction to ≥ **95 %** of maximum dry density in 150–200 mm lifts; use light equipment near the facing.
- For steel strips, backfill should be **non-corrosive**: pH 5–10, resistivity ≥ 3000 Ω·cm, chlorides ≤ 100 ppm, sulphates ≤ 200 ppm, organic content ≤ 1 %.
- Clays, silts, organic soils and frozen soils should be avoided: low friction, poor drainage, pore pressure build-up.

3.2 Reinforcement

- **Metallic strips** – galvanised mild steel ribbed strips, e.g. 50 mm × 4 mm, zinc coating ≈ 70–85 µm. Inextensible, high modulus, high pull-out strength.
- **Steel grids / welded wire mesh / bar mats** – friction + passive resistance on cross bars.
- **Geosynthetics** – geotextiles (woven), geogrids (HDPE, PP, PET), geostrips (polyester yarn in PE sheath). Extensible, non-corrosive, light and easy to handle.
- Ideal properties: high tensile strength, low creep, good soil friction, durable, ductile, easily jointed and cheap.

Note

Corrosion design (galvanised steel, FHWA):

Zinc loss ≈ 15 µm/year for first 2 years, then ≈ 4 µm/year; carbon steel afterwards ≈ 12 µm/year.

Sacrificial thickness is added to the design life (75–100 years for permanent walls).

Extensible vs inextensible reinforcement

Property	Inextensible (steel)	Extensible (geosynthetic)
Strain at failure	< 1–2 %	10 % to 80 %
Modulus	Very high	Low to moderate
Failure surface	Bilinear (coherent gravity)	Rankine plane
Earth pressure coeff.	K0 → Ka with depth (1.7Ka→1.2Ka)	Ka (constant)
Creep	Negligible	Significant for PP / PE
Durability issue	Corrosion	UV, chemical, creep, damage

Failure modes (internal) of reinforced soil

1. **Tension (rupture) failure** – reinforcement breaks; occurs when bond is strong but strength is small. Cure: stronger / closer layers.
2. **Pull-out (slip) failure** – reinforcement slips out of resistant zone; occurs when length L_e is short or friction is low. Cure: longer / rougher reinforcement.
3. **Connection failure** – at facing, due to corrosion or poor detailing.
4. **Excessive deformation** – due to extensible reinforcement, poor compaction.

Design rule: **allowable tension** $\geq T_{max}$ **and pull-out resistance** $\geq T_{max}$ at every layer, with a factor of safety (usually 1.5).

3.3 Facing

- **Precast concrete panels** (cruciform / rectangular, ≈ 1.5 m × 1.5 m, 140–180 mm thick, M30 or higher) – most common for permanent walls.
- **Modular concrete blocks** with geogrid; **gabion** facing; **welded wire** facing with vegetation; **geotextile wrap-around** (temporary / green slopes).
- Facing gives aesthetics and prevents raveling; connection to reinforcement must be strong and corrosion-proof.

4. Soil–Reinforcement Friction Coefficient

4.1 Definition

Reinforced earth works only if the soil can **grip** the reinforcement. This gripping ability is measured by the **coefficient of friction** (μ).

$$\mu = \tau / \sigma_n = \tan\delta$$

τ = interface shear (frictional) stress, σ_n = normal stress on the reinforcement, δ = **angle of interface friction** between soil and reinforcement. The **interaction (efficiency) coefficient** compares it with the soil itself:

$$C_i = \tan\delta / \tan\varphi \ (\leq 1 \text{ for smooth}, \approx 1 \text{ or more for grids})$$

In pull-out analysis the **apparent friction coefficient** μ^* (or F^*) is used, which also includes dilatancy and passive resistance:

$$P_r = 2 \cdot b \cdot L_e \cdot \mu^* \cdot \sigma_v$$

P_r = pull-out resistance, b = strip width, L_e = embedded length in the resistant zone, σ_v = vertical effective stress at the level of the strip, factor 2 = top and bottom faces.

Important

Variation of μ^* with depth (ribbed steel strip, Schlosser):

at top $z = 0$: $\mu^* = 1.2 + \log C_u$ (≤ 2.0) → decreases linearly → at $z = 6$ m and below: $\mu^* = \tan\varphi$.

Default for geosynthetics: $F^* \approx 2/3 \tan\varphi$ (sheets / geogrids).

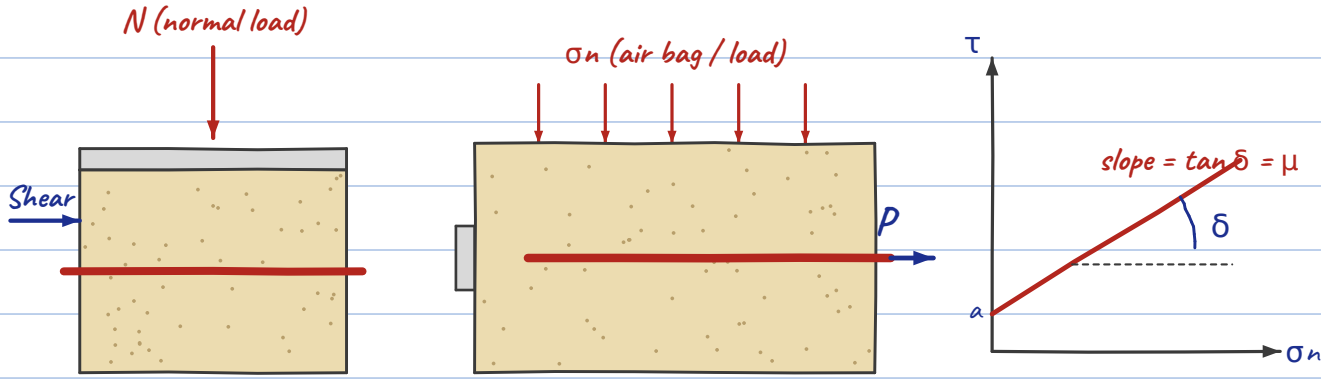
4.2 Laboratory determination

(i) Direct shear test (modified, large shear box – ASTM D5321)

- Lower half of box filled with soil (or rigid block), reinforcement fixed at the interface between the halves, upper half filled with soil compacted to field density.
- Normal load σ_n applied (3–4 values), shear displacement at about 1 mm/min.
- Peak τ plotted against σ_n; **slope = tan δ**, intercept = adhesion a.

(ii) Pull-out test (ASTM D6706) – closest to field behaviour

- Large box with reinforcement embedded in the centre of the soil; a **sleeve** at the front wall removes front-wall effects.
- Normal pressure by rubber air-bag / hydraulic jack; reinforcement pulled out at a constant rate (≈ 1 mm/min).
- Pull-out force P vs displacement recorded; peak P used: $\mu^* = P / (2\,b\,L\,\sigma_n)$.



(a) Direct shear — reinforcement at interface
(b) Pull-out test — strip length L embedded
(c) τ vs σn plot — a = adhesion intercept

Fig 1.3 Laboratory tests for friction coefficient

(iii) Inclined plane (tilt table) test – soil block placed on reinforcement fixed to a plank; plank tilted slowly; at slip angle β, $\delta = \beta$. Used for geosynthetic–soil and geosynthetic–geosynthetic interfaces; simple but low stress levels.

Examples

Example 2 (direct shear): at σ_n = 100 kPa, τ = 62 kPa → μ = 0.62 → $\delta = \tan^{-1}(0.62) = 31.8°$.

Example 3 (pull-out): P = 12 kN, b = 50 mm, L = 1.0 m, σ_n = 100 kPa → $\mu^* = 12 / (2 \times 0.05 \times 1.0 \times 100) = 1.2$

Why is μ^* larger near the top?

A dense granular soil **dilates** when sheared at the interface. The surrounding soil restrains this volume increase, so the normal stress on the strip rises above σ_v. At shallow depth confinement is small and dilatancy is large (μ^* up to 1.5–2); at greater depth the effect vanishes and $\mu^* \to \tan\varphi$.

Interface (typical, indicative)	δ or C_i
Sand – smooth steel strip	$\delta \approx$ ½ to ⅔ φ
Sand – ribbed steel strip	$\mu^* \approx$ 1.2 – 2 (top), tan φ (deep)
Sand – woven geotextile	$C_i \approx$ 0.7 – 0.9
Sand – nonwoven geotextile	$C_i \approx$ 0.8 – 1.0
Sand – geogrid	$C_i \approx$ 0.8 – 1.1

4.3 Factors affecting the friction coefficient

1. **Surface roughness / texture** of reinforcement – ribbed strips and grids give higher μ than smooth strips.
2. **Soil type and gradation** – angular, well-graded sand/gravel gives high φ and μ; fines reduce it.
3. **Particle size vs aperture** (grids) – best interlock when D_{50} is suitable relative to aperture size.
4. **Normal stress** – μ is high at low σ_n (dilatancy) and decreases with depth, approaching tan φ.
5. **Density / degree of compaction** – dense soil dilates more → higher pull-out resistance.
6. **Moisture content / saturation** – wet or saturated fines reduce friction.
7. **Reinforcement stiffness and extensibility** – extensible sheets show progressive mobilisation, so only part of length is active.
8. **Embedded length and width** – scale effects; short strips mobilise more peak μ.
9. **Rate of displacement, box size and boundary conditions** (test factors) affect the measured value.

5. Applications of Reinforced Earth

- **Retaining walls** – highway/railway, flyover and bridge approaches, ports, industrial yards (heights 3–30 m).
- **Bridge abutments** and wing walls (integral and separate).

- **Embankments on soft ground** – basal reinforcement; steep-sided reinforced embankments.
- **Reinforced soil slopes** (RSS) up to 70° – landslide repair, widening of roads in hilly areas.
- **Foundations** – reinforced soil beds under footings, rafts and storage tanks.
- **Roads, railways and airfields** – reinforced base/sub-base on weak subgrade.
- **Dams, levees, sea walls, river training** and coastal protection.
- **Others** – tunnel portals, noise barriers, mine waste dumps, military bunkers, bunds and ramps.

Advantages

- Economical (about 25–50 % cheaper than RCC walls at medium–large heights) and **fast** to build.
- Flexible: tolerates large differential settlement and has good **seismic** performance.
- No formwork, small foundations, uses local granular fill, easy to inspect and repair.
- Aesthetic (various facings) and environment friendly.

Limitations

- Needs good granular backfill and adequate space behind the face ($L \geq 0.7H$).
- Durability of metal (corrosion) / polymers (creep, UV) must be checked.
- Design is sensitive to proper compaction and drainage.

Construction sequence of a reinforced earth wall

1. Excavate, level and compact the foundation; cast unreinforced concrete **leveling pad** (≈ 150 mm × 300 mm).
2. Erect first row of facing panels, plumb and aligned; temporary wedges / bracing.
3. Place and compact backfill in 150–200 mm lifts up to the first reinforcement level (≥ 95 % MDD).

4. Lay reinforcement perpendicular to the face, flat, and connect it to the panels.
5. Spread the next lift; avoid heavy rollers within 1 m of the facing (use hand compactors).
6. Repeat layer-by-layer; provide **drainage** behind and below the reinforced zone; cast coping beam at top.

Formula sheet – Unit I

Quantity	Expression
Interface friction	$\mu = \tau/\sigma_n = \tan\delta$; $C_i = \tan\delta / \tan\varphi$
Pull-out resistance	$P_r = 2\, b\, L_e\, \mu^* \sigma_v$
Enhanced confinement	$\Delta\sigma_3 = T/(S_v S_h)$; $\Delta\sigma_1 = \Delta\sigma_3 K_p$
Pseudo-cohesion	$c_R = (\Delta\sigma_3/2)\sqrt{K_p}$; $K_p = \tan^2(45^\circ + \varphi/2)$
Rankine coefficient	$K_a = \tan^2(45^\circ - \varphi/2) = (1 - \sin\varphi)/(1 + \sin\varphi)$

Important questions

1. Explain the concept of reinforced earth. How is soil strengthened by reinforcement?
2. Explain equal confining pressure and pseudo-cohesion concepts with Mohr circles.
3. Define friction coefficient. Describe pull-out test and direct shear test.
4. List the factors affecting soil–reinforcement friction coefficient.
5. Write briefly on materials and applications of reinforced earth.